

ANUJ K JANARDHANAN

AI/ML Engineer | Generative AI | NLP | Computer Vision
+91 9824350995 | anujanardhanan@gmail.com | Ahmedabad, Gujarat, India
[LinkedIn](#) | [GitHub](#) | [Credly](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with hands on experience building AI systems across semantic search, NLP, computer vision and predictive analytics. Currently developing an AI powered recruitment and semantic search platform using FastAPI, PostgreSQL, pgvector, FastEmbed and asynchronous processing with Redis and Celery. Previously worked on satellite image super resolution research at ISRO's Space Applications Centre and applied machine learning at Adani Enterprises.

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers, FastEmbed

Generative AI / NLP: LLMs, Semantic Search, Information Retrieval, Vector Search, Embeddings, Reranking, Prompt Engineering, Ollama

Deep Learning / CV: CNNs, GANs, Diffusion Models, ESRRGAN, SRDiff, ResNets, SISR, Attention, PSNR / SSIM

Programming: Python, C++, Java, JavaScript, TypeScript

Backend / Data: FastAPI, Node.js, Flask, PostgreSQL, pgvector, MongoDB Atlas, MySQL

Infrastructure / Tools: Docker, Redis, Celery, nginx, Prometheus, Grafana, Git, GitHub, Jupyter, VS Code

WORK EXPERIENCE

Amulya Corporate Services LLP | AI Developer

Jul 2026 - Present

- Engineering AMDEX, an AI powered recruitment and semantic candidate search platform (in development/testing) built on structured candidate data and semantic representations, designed to scale to a 5,000+ resume corpus.
- Implemented hybrid candidate retrieval combining PostgreSQL full text search, skill matching and pgvector semantic similarity, followed by cross encoder reranking. Developed JD based matching workflows that interpret job requirements and rank candidates using semantic and structured matching signals.
- Built an asynchronous resume processing pipeline using FastAPI, PostgreSQL, Redis and Celery.
- Containerized the application with Docker and nginx, with Prometheus and Grafana monitoring for deployment and testing.

Space Applications Centre, ISRO | Project Trainee - Deep Learning & Computer Vision

Dec 2025 - Jun 2026

- Built ESRRGAN and SRDiff super resolution models from scratch in PyTorch to improve the clarity of low resolution satellite images, tuning perceptual loss functions and measuring output quality with PSNR and SSIM.
- Worked on SRDiff, a diffusion based architecture that uses stochastic denoising to recover fine image detail that standard upscaling methods miss.
- Owned the full training pipeline data preprocessing, CNN architecture with attention and multi scale feature extraction, training loops and iterative debugging working independently end to end.

Adani Enterprises Limited | Analytics & AI Intern

Jul 2025 - Sep 2025

- Built a churn prediction model for the ICM Project using XGBoost and K-Means clustering to flag at risk customers, delivering AUC-ROC evaluated churn probability scores to the analytics team.
- Ran the complete ML workflow in Python: cleaned raw data, engineered features, tuned hyperparameters and translated model outputs into plain business recommendations for retention planning.

PROJECTS

WorkSync AI: Intelligent HR Management System | React, TypeScript, Node.js, MongoDB Atlas, Tailwind CSS, Ollama

- Built a chat assistant on top of a local LLM (Ollama) that lets HR staff query employee data in plain English no SQL needed and wired it into a React/TypeScript frontend with Tailwind CSS.
- Wrote RESTful APIs in Node.js backed by MongoDB Atlas, handled real time data operations, access control and end to end integration across the full stack.

[GITHUB](#)

Moodify: AI Emotion Detection Wellness App | Python, Flask, Scikit-learn, NLP, Ollama, Generative AI

- Trained an NLP classifier to detect user emotion from free text input and map it to a mood persona, plugged in a local generative AI model (Ollama) for follow up conversation with prompt engineered responses.
- Built the full app in Flask user auth, session handling and a UI that changes dynamically based on the detected mood, kept the architecture modular so a fine tuned transformer can slot in later.

[GITHUB](#)

Weekendy: Hotel Booking & Itinerary Platform | JSP, Servlets, HTML, CSS, Apache Derby (JDBC)

- Designed a hotel booking site with separate user and admin roles, live MySQL updates and a clean responsive interface focused on making the flow simple and the data always in sync.
- Built end to end CRUD booking workflows using JDBC and prepared statements, including date range validation and real time room availability checks

[GITHUB](#)

EDUCATION

Indus University, Ahmedabad - B.Tech Computer Science Engineering

2022 – 2026

CGPA: **9.16 / 10** | Relevant Coursework: Machine Learning, OOPS, Data Structures & Algorithms, Database Systems

Shree Narayana Guru Vidhyalaya - Secondary & Higher Secondary Education

2010 – 2022

CERTIFICATIONS & ACHIEVEMENTS

AWS Academy Machine Learning Foundations - Amazon Web Services (AWS Academy) | Aug 2025 | 20 hrs | [View Credly Badge](#)

Data Analysis Using Python – IBM (Credly) | June 2026 | [View Credly Badge](#)

ISRO Project Trainee - Space Application Centre - Competitively selected for national level deep learning research program on satellite image super resolution.